Lecture 12 : Stochastics

Suppose that, given some data, we want to "learn" the function

$$F: \text{recent data} \longrightarrow \text{future}$$

which will help us with weather forecasting.

We build a Neural Network (NN) that finds $\hat{F}$ s.t. $\hat{F} \approx F$.

We perform supervised learning : we gather enough pairs (x_{recent}, x_{future}), check how good our predictions are (error of $\hat{F}$ approximation) and then modify $\hat{F}$ accordingly.

Problem: Measurements are not exact $\Rightarrow$ existence of noise (random errors) in our observations

$(x_{recent} + error_1 , x_{recent} + error_2)$

we don't have exact learning data.

$\Downarrow$

For thousands of observations, the noise may worsened NN predictions.

$\Leftarrow$ Use probability methods instead.

With enough data we can recover a good approximation of our relation.

Random variables

$$X: \Omega \longrightarrow \mathbb{R}$$

Where Ω is the set of all possible outcomes of an experiment.
↳ sample space

- Example: $\Omega = \{\omega / \omega \text{ is a number from 1 to 6 (dice)}\}$. We perform a random experiment : roll a dice $\Rightarrow$ This produces an outcome $\omega \in \Omega$. Then the random variable $X: \Omega \to \mathbb{R}$ assigns a number to this outcome (actually to all outcomes): $X(\omega) = \begin{cases} 1, & \omega \text{ even.} \\ 0, & \omega \text{ odd.} \end{cases}$

We call $A \subseteq \Omega$ an event, e.g. $A = \{1, 3, 5\}$. To each event, we will

assign it a probability $P(A)$: this measures how likely is this to happen.

Properties:

1. If $A_1 \cap A_2 = \phi$ (disjoint events) $\Rightarrow$ $P(A_1 \cup A_2) = P(A_1) + P(A_2)$
2. $P(\Omega) = 1$.

There are two kinds of random variables X : discrete and continuous.

This is mainly due to the issue that If we would like to assign a probability to an event such as :

$$A = \{ \omega \,/\, X(\omega) = 183.4\text{cm} \}$$

(event A represents picking someone from a population that is 183.4cm tall exactly).

$$\Rightarrow P(A) = 0$$

Here X is a continuous random variable, its domain Ω is uncountably infinite.

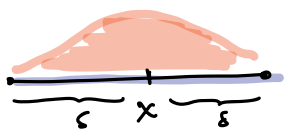
Nonetheless, we can assign probabilities to intervals :

$$A = \{ \omega / \ 160\,\text{cm} \leq X(\omega) \leq 180\,\text{cm} \}$$

and $P(A)$ is not necessarily zero.

We can assign a density function to X :

$$f_X(x) = \lim_{\delta \to 0} \frac{P(x-\delta < X \leq x+\delta)}{2\delta}$$



f_X measures how "dense" the probability is around x.

Then we say X is continuously distributed and we can compute the probability of events such as : $E = \{ \omega / \ a < X(\omega) < b \} \subseteq \Omega$

in the following way:

$$P_X(a < x < b) = \int_a^b f_X(x)\,dx$$

Some important examples

We pick a random variable $X : \Omega \to \mathbb{R}$ uniformly distributed on $[a, b]$.

This tells me $X(\Omega) \subseteq [a, b]$ with density function: $f_X(x) = \begin{cases} \frac{1}{b-a} & ; a \leq x \leq b, \\ 0 & ; \text{otherwise.} \end{cases}$

Notation: $X \sim U(a,b)$

In this distribution, every interval of the same length in $[a,b]$ is equally likely.

Then:

$$P(l_1 < x \, l_2) = \int_{l_1}^{l_2} f_x(x)\,dx$$

$$= \int_{l_1}^{l_2} \frac{1}{b-a}\,dx = \frac{l_2 - l_1}{b-a}.$$

Expected value: It's a probability weighted average of all possible values of the random variable:

$$\mathbb{E}[x] := \int_{-\infty}^{\infty} x \cdot f_x(x)\,dx \rightarrow \text{For } X \sim U(a,b): \mathbb{E}[x] = \frac{a+b}{2}$$

That is, suppose we perform a random experiment ∞-many times. The expected value tells me what would be the average of the values that we observe.

Law of Large Numbers*

$$\frac{X(w_1) + X(w_2) + \dots + X(w_N)}{N} \xrightarrow{N \to \infty} \mathbb{E}[x]$$

*($w_1, w_2, \dots, w_N$ are independent repetitions of the same experiment)

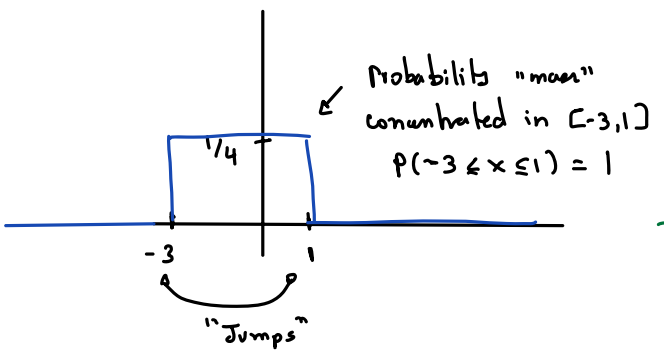
Variance: How far spread are our observed values from the expected value:

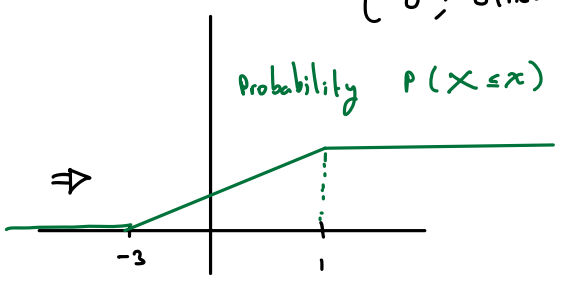
$$\left(X(w_i) - \mathbb{E}[X]\right)^2 \quad \text{deviation of observation } i$$

We define the variance as the average of all these deviations, hence we take $\mathbb{E}[\cdot]$:

$$Var(X) = \mathbb{E}\left(X - N\right)^2. \rightarrow \text{For } X \sim U(a,b): Var(X) = \frac{(b-a)^2}{12}.$$

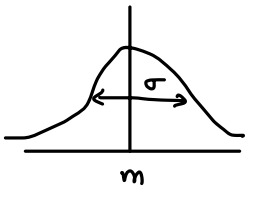
Example: A uniform density function ①: $X \sim U(-3,1)$. Then $f_x(x) = \begin{cases} \frac{1}{4} & ; x \in [-3,1) \\ 0 & ; \text{otherwise.} \end{cases}$





Normal distribution

$X \sim N(m, \sigma)$ — m: mean; σ: standard deviation



Here the density function is

$$f_{m,\sigma}(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-m)^2}{2\sigma^2}}$$

Properties:

- Expected value : m, Variance : σ^2. → If $m=0$, $\sigma^2=1$ we say X has a standard normal distribution.

- It is useful for simulating random noise in deterministic data: data = observed data + noise where noise $\sim N(0,\sigma^2)$.

↓

Roughly speaking, since error can be produced by a lot of different factors, when we add these factors together, these independent factors will tend to a normal distribution.

↳ Central limit theorem: Suppose $X_1, \dots, X_n$ are random variables independent with the same distribution and $\mathbb{E}[X_i] = N$, $\mathrm{Var}[X_i] = \sigma^2$. Then

$$\bar{X} := \frac{1}{n}(X_1 + \dots X_n)$$

satisfies

$$\frac{\bar{X} - N}{\sigma^2/\sqrt{n}} \sim N(0,1) \quad \left(\text{equivalently } \bar{X} \approx N\left(N, \frac{\sigma^2}{n}\right)\right)$$

Remark: Adding random noise to our data can help us test the robustness of a method: $F(x) + F(x+\varepsilon)$ should be similar for small ε.

Transformation of random variables

Basic idea: if X is a random variable, then so is $g(X)$.

Example: $X \sim U(a,b)$ and $k>0$, Then:

(shifting) $g(x) = X + k \sim U(a+k, b+k)$

(scaling) $g(x) = kX \sim U(ka, kb)$

Be careful: if $X_1, X_2 \sim U(a,b)$ are independent, it is not true that $X_1+X_2 \sim U(2a, 2b)$.

Normal distribution: $X \sim N(0,1)$ and $k>0$:

(shifting) $g(x) = X + k \sim N(k,1)$, i.e. $f_{g(x)}(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{(x-k)^2}{2}}$

Also notice that for m, σ:

$$g(x) = m + \sigma X \sim N(m, \sigma^2) \quad \text{i.e.} \quad f_{g(x)}(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-m)^2}{2\sigma^2}}$$

We can transform it back! (Standarizing normal variable)

That is, from $X \sim N(m, \sigma^2)$ to $\tilde{X} \sim N(0,1)$

How?

$$\text{Set} \quad \tilde{X} = \frac{X - m}{\sigma}.$$

Remark: For the normal distribution if X_1, X_2 are independent, normally distributed, then $X_1 + X_2$ is also normally distributed.

Is there a way to generate random data that is distributed in a specific way? Answer: Yes!

For example, generate $y_1, y_2, \ldots, y_n \sim U(0,1)$ and use them to obtain

$$x_1, x_2, \ldots, x_n$$

with a specific distribution, say with density function $f_X(x)$. We define its Cumulative Distribution function: $F_X(x) = P(X \leq x) = \int_{-\infty}^{x} f_X(x)dx$.

Then we set $x_1 = F_X^{-1}(y_1)$, $x_2 = F_X^{-1}(y_2)$, $\ldots$, $x_n = F_X^{-1}(y_n)$.

We can show that $x_1, \dots, x_n$ have the same CDF:

$$P(x_i \le x) = P\left(F_x^{-1}(y_i) \le x\right)$$

↓ F_x is increasing

$$= P\left(y_i \le F_x(x)\right)$$

We know that $y_i \sim U(0,1)$, i.e. $P(y_i \le a) = \int_0^a 1\,dx = a$. Then

$$P(y_i \le F_x(x)) = F_x(x)$$

$$\Rightarrow \quad P(x_i \le x) = F_x(x). \qquad \blacksquare$$

Standardization of random variables

Motivation: We can think of our data as

$$\text{data} = \text{"real value"} + \underbrace{\text{noise}}_{\sim N(0,\sigma^2)} \qquad \sim N(\mu, \sigma^2)$$

but we don't know these parameters μ, σ a priori. What we can do is to use their estimates.

$\Rightarrow$ Instead of $x_1, x_2, \dots, x_n$ we use

$$\frac{x_1 - \bar{x}}{\sigma_x}, \frac{x_2 - \bar{x}}{\sigma_x}, \dots, \frac{x_n - \bar{x}}{\sigma_x}$$

where $\bar{x}$ → average and σ_x – standard deviation

$$\bar{x} = \frac{1}{n}\sum x_i \qquad \sigma_x^2 = \frac{1}{n}\sum (x_i - \bar{x})^2$$

} sample mean and sample standard dev.

- Another option: Min Max Scaler in Python. We transform $x_1, x_2, \dots, x_n$ into:

$$\frac{x_1 - \bar{x}}{\max\limits_j x_j - \min\limits_j x_j}, \frac{x_2 - \bar{x}}{\max\limits_j x_j - \min\limits_j x_j}, \dots, \frac{x_n - \bar{x}}{\max\limits_j x_j - \min\limits_j x_j}$$

In real-life situations, these are still not enough: a prefiltering of data is necessary. (min and max are affected by both extremes of data).

Vector-valued random variables

In practice, we measure more quantities for the same observation, e.g.,

$$\rightsquigarrow W = \begin{pmatrix} T \\ h \\ p \\ w \\ \vdots \end{pmatrix} \quad \begin{matrix} \text{Temperature.} \\ \text{humidity.} \\ \text{pressure.} \\ \text{wind speed} \\ \vdots \end{matrix}$$

weather

An important case: Two random variables X, Y follow the bivariate normal dist. if

$$aX + bY \sim N(\mu, \sigma^2) \quad \text{for any } a, b \in \mathbb{R}.$$

Example: Calorie intake and weight of an individual in some population:

observed variables: $\underbrace{(\text{energy}, \text{weight})}_{\text{2 components}}$

How to describe such a distribution?

We collect each mean: $\mu = \begin{pmatrix} \mathbb{E}[X] \\ \mathbb{E}[Y] \end{pmatrix} = \begin{pmatrix} \mu_X \\ \mu_Y \end{pmatrix}$.

variance / covariance matrix: $\Sigma = \begin{pmatrix} \text{Var}(X) & \text{Cov}(X,Y) \\ \text{Cov}(X,Y) & \text{Var}(Y) \end{pmatrix}$

$\text{Cov}(X,Y) = \mathbb{E}[(X-\mu_X)(Y-\mu_Y)]$

Bivariate Normal distribution

Density function: $f : \mathbb{R}^2 \to \mathbb{R}$

$$x \longmapsto f(x) = \frac{1}{2\pi\sqrt{\det(\Sigma)}} \exp\left\{ \underbrace{-\frac{1}{2}(x-\mu)^T \Sigma^{-1} (x-\mu)}_{\text{"}\Sigma^{-1}\text{-Distance" of } x \text{ to the mean } \mu} \right\}$$

$\sqrt{\det(\Sigma)}$: scaling of the overall area transformed by Σ.

This constant normalizes f

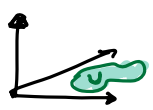
↳ $\|x-y\|^2_{\Sigma^{-1}} := (x-y)^T \Sigma^{-1} (x-y)$

Recall:

$\Sigma \to$ covariance matrix.

$\mu \to$ mean vector.

Then $P(x \in U) = \int_U f(p)\, dp$

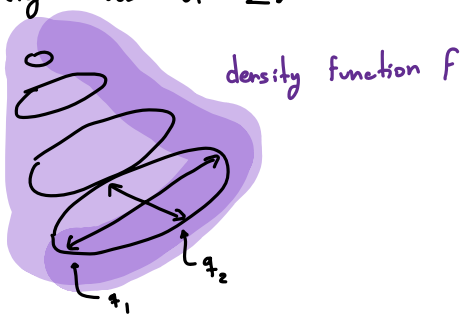


Remark: Since Σ is symmetric $\Rightarrow$ eigenvectors form an orthonormal basis:

That is, we have $\{q_1, q_2\}$ eigenvectors s.t. $\underbrace{q_i \cdot q_j}_{q_j^T q_i} = \delta_{ij}$. Thus we know we can diagonalize Σ using eigenvalue decomposition:

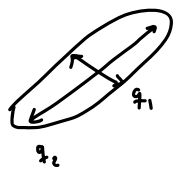
$$\Sigma = Q D Q^T \quad \text{where} \quad Q = \begin{pmatrix} q_1 & q_2 \end{pmatrix} \text{ and } D = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}$$

where λ_1, λ_2 are eigenvalues of Σ.



Then Σ (and Σ^{-1}) can be seen as a rotation/reflection, followed by a scaling given by the eigenvalue of Σ. This will affect the shape of our "bell" function f.

$\Rightarrow$ q_1, q_2 tell me the principal directions of the ellipse and eigenvalues how much data would "spread out" in these directions.

e.g. 

large λ_2 but small λ_1.

This is easier to see by studying the level curves of f and noticing they are of the form:

$$\frac{z_1^2}{\lambda_1} + \frac{z_2^2}{\lambda_2} = c.$$

where z_1, z_2 can be understood as the coordinates of $(x-\mu)$ in the $\{q_1, q_2\}$ basis.

$\Rightarrow$ $\sqrt{\lambda_1}, \sqrt{\lambda_2}$ can be understood as the standard deviation of each coordinate.